

Nilesh S. Patil

Full Stack Developer | GenAI Engineer | Mobile App Developer

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- 8x Hackathon Winner
- IIT Bombay Intern
- Rs.2L Funded Founder
- NASA SpaceApps 1st Place (Local)

PROFILE

Third-year B.E. (AI & DS) student and serial builder with **3+ years** of production development experience. Co-founded **MediAssist AI** (Rs.2L funded by Government of Karnataka NAIN 2.0), shipped **triponbuddy.com** serving live users, and won **8 hackathon competitions** including GDG and NASA SpaceApps 1st place. Proficient across the full stack: React, Next.js, Node.js, Python (Flask/FastAPI), Flutter, and cloud-native infrastructure. Deep experience in **Agentic AI, RAG pipelines, and MCP**; passionate about building AI-native products at the intersection of LLMs, blockchain, and mobile.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Dart, Kotlin, C/C++, Go, SQL, Solidity
Frontend	React, Next.js, HTML5, CSS3, Tailwind CSS, Flutter, ReactFlow, Vite
Backend	Node.js, NestJS, FastAPI, Flask, Express.js
Databases & ORM	Supabase, PostgreSQL, Cloud Firestore, Firebase, Prisma, Redis
AI / ML	Gemini AI, OpenAI API, RAG, Agentic AI, MCP, LangChain, Google ML Kit, Computer Vision
Infrastructure	Docker, GitHub Actions, Vercel, Render, Swagger/OpenAPI, JWT
Blockchain	Solidity, Ethereum, Hardhat, Web3.js

EXPERIENCE

Intern

IIT Bombay (IITB)

2024

- Completed an internship at IIT Bombay, one of India's premier technical institutes.

Freelance Full Stack Developer

Independent Clients (India, USA, Japan)

2023 – Present

- Delivered production applications for clients across domains: real estate diagnostics (Next.js + TypeScript, large-scale codebase), interior design firms (Next.js + Tailwind), and Japanese restaurant booking automation (Go, custom TLS fingerprinting, proxy networking).
- Shipped e-commerce platforms (Next.js + Supabase), event management systems (PHP + TypeScript), and enterprise price-matching tools (Next.js + Supabase + GitHub Actions CI/CD).

PROJECTS

TripOnBuddy: AI Travel Planning Platform

triponbuddy.com

JavaScript · React · Node.js · Python · Supabase · Gemini AI · Docker · Vercel

- Developed a full-stack AI travel platform generating personalized itineraries via Gemini AI based on user preferences, budget, and travel style; live with real users.
- Set up a Node.js + Express backend with Supabase (PostgreSQL) for user data, bookings, and session management; deployed on Vercel + Docker.
- Launched an SEO pipeline (sitemap, robots.txt, Google Search Console) driving organic traffic growth.

MediAssist AI: Dual-App Healthcare Platform (Patient + Doctor)

Rs.2L Funded · Govt. of Karnataka NAIN 2.0

Founder & Project Lead | Flutter · Dart · Supabase · OpenAI GPT-4o · Whisper · OpenAI TTS · Riverpod · Cloud Firestore · Google Maps

- Co-founded and architected a full-featured dual-app healthcare system (patient app + doctor provider app) in Flutter/Dart, sharing a unified Supabase backend (PostgreSQL, Auth, Storage, Edge Functions, RLS policies).
- Engineered an AI voice assistant with OpenAI Whisper (speech-to-text), GPT-4o (conversational health AI with session context), and OpenAI TTS (voice-back responses); full voice-to-voice multi-modal loop.
- Integrated prescription scanning (camera OCR + GPT-4o Vision), medication reminders with native push notifications, medical records management, vital tracking, and Emergency SOS alerts to saved contacts.
- Doctor portal: appointment calendar management, patient history, clinical notes, real-time Accept/Decline via Supabase live subscriptions, and multilingual regional language support for elderly and rural demographics.

College ERP System: Production-Grade Enterprise Application

NestJS · TypeScript · React · Prisma · PostgreSQL · Docker · Redis · MinIO · JWT · Swagger · Jest · Playwright

- Developed a production college management system with NestJS + TypeScript backend, Prisma ORM, and PostgreSQL; full RBAC with JWT (Admin / Faculty / Student roles).
- Structured a Docker Compose architecture orchestrating PostgreSQL, Redis, and MinIO; auto-generated Swagger/OpenAPI docs; E2E tests with Playwright and unit tests with Jest.
- Delivered CSV bulk import/export, audit logging, attendance and marks tracking; deployed on Render + Vercel.

Agentic Commerce Protocol: AI Blockchain Payment System

[GitHub](#)

Solidity · Ethereum · Hardhat · Gemini AI · JavaScript · Python · Agentic AI

- Architected a blockchain payment protocol enabling AI agents to transact autonomously via Solidity smart contracts with restricted session keys, spending limits, and verified merchant allowlists.
- Powered by Gemini AI for agentic purchase analysis and budget optimization; built with Hardhat for contract deployment and testing on Ethereum.

AI Social Media Automation: Multi-Platform Content Pipeline

Python · Flask · Gemini AI · Ideogram v2 · Instagram API · LinkedIn API · Twitter API · YouTube API

- Shipped a full agentic AI content pipeline: Gemini Pro generates prompts, Ideogram v2 creates ad images, Gemini Vision scores quality (auto-publishes if score ≥7.5/10, otherwise loops EDIT or REGENERATE).
- Integrated Instagram Graph, LinkedIn, Twitter, and YouTube APIs for cross-platform automated posting; supports batch Excel uploads, scheduling, and multi-user auth.

Yukti-AI: Visual AI Science Experiment Lab

[GitHub](#)

React · TypeScript · ReactFlow · Vite · Gemini AI · OpenAI · Zustand

- Created a drag-and-drop visual experiment builder with React + ReactFlow, featuring 1000+ components across electronics, chemistry, physics, biology, and quantum domains.
- Integrated Gemini AI + OpenAI for experiment outcome prediction, real-time analysis, and AI-powered suggestions; supports JSON import/export.

RoofVision AI: Satellite Solar Panel Detection

[Live Demo](#)

Python · Flask · Gemini Vision · Google Maps API · Geocoding API · pandas · Render

- Engineered an automated satellite imagery pipeline that geocodes CSV address lists via Google Maps Geocoding API, fetches overhead imagery, and runs Gemini Vision AI for solar panel detection with confidence scoring.
- Outputs structured CSV reports (verdict, confidence %, details, timestamp); deployed on Render; processes up to 25K API calls/day within free tier quotas.

Detox AI: Android Digital Wellness Launcher

[GitHub \(Private\)](#)

Flutter · Dart · Kotlin · Gemini 2.5 Flash · Android AccessibilityService · UsageStatsAPI · Hive · ForegroundService

- Developed a full Android home screen replacement enforcing daily screen time limits with monochrome UI, usage tracking via AccessibilityService + UsageStatsAPI, ForegroundService for persistence.
- Created an agentic AI task-verification loop using Gemini 2.5 Flash multi-modal AI where users earn phone minutes by completing real-world tasks (walking, reading, creative work); AI autonomously verifies photo/video/text submissions.

ACHIEVEMENTS

- 8x Hackathon Competition Winner: GDG (Google Developer Group), CodeBharat (Rs.50,000 prize), ONEST (Rs.25,000 prize), and others.
- 1st Place: NASA SpaceApps Challenge (Local Round)
- Rs.2 Lakh Government Funding: Secured NAIN 2.0 grant from Government of Karnataka as Founder of MediAssist AI.
- Intern at IIT Bombay: Completed internship at India's top engineering institute.
- Shipping live products: TripOnBuddy (AI travel planning), Unyfiny, CMN Services India, and Prasan Hom are among client projects currently live in production.

EDUCATION

S.G. Balekundari Institute of Technology
B.E. in Artificial Intelligence & Data Science

2023 – 2027